

Course No. Math - 203

Course Name : complex variables
& Transforms

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Book References :

1. Advanced Engineering Mathematics
— By H.K Das

2. Complex Analysis
— By Abdul Kuddus.

Complex Number System

Complex Numbers:

The number having the form $x+iy$ is called complex number, where x and y are real numbers, and i which is imaginary unit has the property $i^2 = -1$ (or $i = \sqrt{-1}$).

If $z = x+iy$ be a complex number, then x is called the real part of z and is denoted by $\operatorname{Re}\{z\}$, and y is called imaginary part of z and it is denoted by $\operatorname{Im}\{z\}$.

For example:

1. If $z = 3+2i$ then

$$\operatorname{Re}\{z\} = 3 \text{ and } \operatorname{Im}\{z\} = 2.$$

2. If $z = 2 - \frac{3}{2}i$ then

$$\operatorname{Re}\{z\} = 2 \text{ and } \operatorname{Im}\{z\} = -\frac{3}{2}.$$

3. If $z = -\frac{3}{2} + 4i$ then

$$\operatorname{Re}\{z\} = -\frac{3}{2} \text{ and } \operatorname{Im}\{z\} = 4.$$

we can also define a complex number $x+iy$ as an ordered pair (x, y) of real numbers x and y such that $(x, y) = x+iy = z$.

for example:

$$1. (2, 3) = 2+3i = z.$$

$$2. (-2, 3) = -2+3i = z.$$

$$3. (2, -3) = 2-3i = z.$$

$$4. (-2, -3) = -2-3i = z.$$

$$5. (0, 2) = 0+2i = 2i = z.$$

Complex conjugate:

If $z = x+iy$ be a complex number, then the complex conjugate of z is denoted by $\bar{z}$ or z^* , and is defined by

$$\bar{z} = \overline{x+iy} = x-iy.$$

for example:

$$1. \text{ If } z = 3+2i, \text{ then } \bar{z} = 3-2i.$$

$$2. \text{ If } z = 2-3i, \text{ then } \bar{z} = 2+3i.$$

$$3. \text{ If } z = 4i, \text{ then } \bar{z} = -4i.$$

$$4. \text{ If } z = 5, \text{ then } \bar{z} = 5.$$

Complex number in polar form:

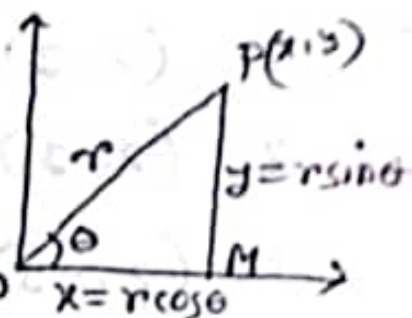
We know, the complex number in Cartesian form is $z = x + iy$ — (1)

putting $x = r \cos \theta$ and $y = r \sin \theta$ in (1), we get

$$z = r \cos \theta + i r \sin \theta$$

$$= r (\cos \theta + i \sin \theta)$$

$$= r e^{i\theta} \quad [\because e^{i\theta} = \cos \theta + i \sin \theta]$$



$\therefore \boxed{z = r e^{i\theta}}$ which is called complex number in polar form.

More than one complex numbers in Cartesian form:

$$z_1 = x_1 + iy_1$$

$$z_2 = x_2 + iy_2$$

$$\text{-----}$$

$$z_n = x_n + iy_n$$

More than one complex numbers in polar form:

$$z_1 = r_1 e^{i\theta_1}$$

$$z_2 = r_2 e^{i\theta_2}$$

$$\text{-----}$$

$$z_n = r_n e^{i\theta_n}$$

Modulus or absolute value:

If $z = x + iy$ be a complex number, then the modulus or absolute value of z is denoted by $|z|$ or r , and is defined by $|z| = r = \sqrt{x^2 + y^2}$.

For example:

1. If $z = 3 + 4i$, then modulus of z is

$$|z| = r = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5.$$

2. If $z = 3 - 4i$, then modulus of z is

$$|z| = r = \sqrt{3^2 + (-4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5.$$

Argument or amplitude:

If $z = x + iy$ be a complex number, then the argument or amplitude of z is denoted by $\arg(z)$ or $\text{amp}(z)$ or θ , and is defined by

$$\arg(z) = \theta = \tan^{-1} \frac{y}{x}.$$

For example:

If $z = 2 + 2\sqrt{3}i$, then argument of z is

$$\arg(z) = \theta = \tan^{-1} \frac{2\sqrt{3}}{2} = \tan^{-1} \sqrt{3} = \tan^{-1} \tan 60^\circ = 60^\circ.$$

problem:

prove that the following properties:

(a) $\overline{\overline{z}} = z$ (b) $z + \overline{z} = 2 \operatorname{Re}\{z\}$ (c) $z\overline{z} = |z|^2$

(d) $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$ (e) $\overline{z_1 z_2} = \overline{z_1} \overline{z_2}$.

Solution:

(a) Let $z = x + iy$

$$\therefore \overline{z} = x - iy$$

$$\therefore \overline{\overline{z}} = x + iy$$

$$\Rightarrow \overline{\overline{z}} = z \text{ (proved)}$$

(b) Let $z = x + iy \therefore \overline{z} = x - iy$

$$L.H.S = z + \overline{z}$$

$$= x + iy + x - iy$$

$$= 2x$$

$$= 2 \operatorname{Re}\{z\} \quad [\because z = x + iy, \operatorname{Re}\{z\} = x]$$

$$= R.H.S$$

$$\therefore L.H.S = R.H.S \text{ (proved)}$$

(c) Let $z = x + iy \therefore \overline{z} = x - iy$

$$L.H.S = z\overline{z}$$

$$= (x + iy)(x - iy)$$

$$= x^2 - (iy)^2$$

$$= x^2 - i^2 y^2$$

$$= x^2 - (-1)y^2 \quad (\because i^2 = -1)$$

$$= x^2 + y^2$$

$$L.H.S = x^2 + y^2$$

$$= |z|^2 \quad [\because |z| = \sqrt{x^2 + y^2} \Rightarrow |z|^2 = x^2 + y^2]$$

$$= R.H.S$$

$$\therefore L.H.S = R.H.S \text{ (proved)}$$

$$(d) \quad \overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$$

$$\text{Let } z_1 = x_1 + iy_1 \quad \therefore \overline{z_1} = x_1 - iy_1$$

$$\text{and } z_2 = x_2 + iy_2 \quad \therefore \overline{z_2} = x_2 - iy_2$$

$$\cancel{L.H.S} = \cancel{z_1 + z_2}$$

$$\text{Now, } z_1 + z_2 = x_1 + iy_1 + x_2 + iy_2$$

$$\Rightarrow z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2)$$

$$\therefore \overline{z_1 + z_2} = (x_1 + x_2) - i(y_1 + y_2)$$

$$= (x_1 - iy_1) + (x_2 - iy_2)$$

$$\Rightarrow \overline{z_1 + z_2} = \overline{z_1} + \overline{z_2} \text{ (proved)}$$

$$(c) \overline{z_1 z_2} = \overline{z_1} \overline{z_2}$$

Let in polar form,

$$z_1 = r_1 e^{i\theta_1} \quad \therefore \overline{z_1} = r_1 e^{-i\theta_1}$$

$$\text{and } z_2 = r_2 e^{i\theta_2} \quad \therefore \overline{z_2} = r_2 e^{-i\theta_2}$$

$$\text{Now, } z_1 z_2 = r_1 e^{i\theta_1} \cdot r_2 e^{i\theta_2}$$

$$= r_1 r_2 \cdot e^{i\theta_1} \cdot e^{i\theta_2}$$

$$= r_1 r_2 e^{i(\theta_1 + \theta_2)}$$

$$= r_1 r_2 e^{i(\theta_1 + \theta_2)}$$

$$\therefore \overline{z_1 z_2} = r_1 r_2 e^{-i(\theta_1 + \theta_2)}$$

$$= r_1 r_2 e^{-i\theta_1} \cdot e^{-i\theta_2}$$

$$= r_1 e^{-i\theta_1} \cdot r_2 e^{-i\theta_2}$$

$$\Rightarrow \overline{z_1 z_2} = \overline{z_1} \overline{z_2} \text{ (proved)}$$

problem:

For any two complex numbers z_1 and z_2 prove that

$$(a) \arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$$

$$(b) \arg\left(\frac{z_1}{z_2}\right) = \arg(z_1) - \arg(z_2)$$

$$(c) |z_1 z_2| = |z_1| |z_2|$$

$$(d) \left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$$

Solution:

$$(a) \arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$$

let in polar form,

$$z_1 = r_1 e^{i\theta_1} \quad \therefore \arg(z_1) = \theta_1$$

$$\text{and } z_2 = r_2 e^{i\theta_2} \quad \therefore \arg(z_2) = \theta_2$$

$$\text{Now, } z_1 z_2 = r_1 e^{i\theta_1} \cdot r_2 e^{i\theta_2}$$

$$= r_1 r_2 e^{i\theta_1} \cdot e^{i\theta_2}$$

$$= r_1 r_2 e^{i\theta_1 + i\theta_2}$$

$$= r_1 r_2 e^{i(\theta_1 + \theta_2)}$$

$$\therefore \arg(z_1 z_2) = \theta_1 + \theta_2$$

$$\Rightarrow \arg(z_1 z_2) = \arg(z_1) + \arg(z_2) \text{ (proved)}$$

$$(b) \arg\left(\frac{z_1}{z_2}\right) = \arg(z_1) - \arg(z_2)$$

let in polar form,

$$z_1 = r_1 e^{i\theta_1} \quad \therefore \arg(z_1) = \theta_1$$

$$\text{and } z_2 = r_2 e^{i\theta_2} \quad \therefore \arg(z_2) = \theta_2$$

$$\text{Now, } \frac{z_1}{z_2} = \frac{r_1 e^{i\theta_1}}{r_2 e^{i\theta_2}}$$

$$= \frac{r_1}{r_2} e^{i\theta_1 - i\theta_2}$$

$$= \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}$$

$$\therefore \arg\left(\frac{z_1}{z_2}\right) = \theta_1 - \theta_2 = \arg(z_1) - \arg(z_2) \text{ (proved)}$$

$$(c) |z_1 z_2| = |z_1| |z_2|$$

In polar form,

$$\text{let } z_1 = r_1 e^{i\theta_1} \therefore |z_1| = r_1$$

$$\text{and } z_2 = r_2 e^{i\theta_2} \therefore |z_2| = r_2$$

$$\text{Now, } z_1 z_2 = r_1 e^{i\theta_1} \cdot r_2 e^{i\theta_2}$$

$$= r_1 r_2 e^{i\theta_1} \cdot e^{i\theta_2}$$

$$= r_1 r_2 e^{i\theta_1 + i\theta_2}$$

$$= r_1 r_2 e^{i(\theta_1 + \theta_2)}$$

$$\therefore |z_1 z_2| = r_1 r_2$$

$$\Rightarrow |z_1 z_2| = |z_1| |z_2| \text{ (proved)}$$

$$(d) \left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$$

In polar form,

$$\text{let } z_1 = r_1 e^{i\theta_1} \therefore |z_1| = r_1$$

$$\text{and } z_2 = r_2 e^{i\theta_2} \therefore |z_2| = r_2$$

$$\text{Now, } \frac{z_1}{z_2} = \frac{r_1 e^{i\theta_1}}{r_2 e^{i\theta_2}}$$

$$= \frac{r_1}{r_2} e^{i\theta_1 - i\theta_2}$$

$$= \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}$$

$$\therefore \left| \frac{z_1}{z_2} \right| = \frac{r_1}{r_2} = \frac{|z_1|}{|z_2|} \text{ (proved)}$$

problems:

prove that $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2|z_1|^2 + 2|z_2|^2$

solution:

Let $z_1 = x_1 + iy_1$

and $z_2 = x_2 + iy_2$

$$\therefore |z_1| = \sqrt{x_1^2 + y_1^2}$$

$$\therefore |z_2| = \sqrt{x_2^2 + y_2^2}$$

$$\Rightarrow |z_1|^2 = x_1^2 + y_1^2$$

$$\Rightarrow |z_2|^2 = x_2^2 + y_2^2$$

$$L.H.S = |z_1 + z_2|^2 + |z_1 - z_2|^2$$

$$= |x_1 + iy_1 + x_2 + iy_2|^2 + |x_1 + iy_1 - (x_2 + iy_2)|^2$$

$$= |(x_1 + x_2) + i(y_1 + y_2)|^2 + |(x_1 - x_2) + i(y_1 - y_2)|^2$$

$$= \left\{ \sqrt{(x_1 + x_2)^2 + (y_1 + y_2)^2} \right\}^2 + \left\{ \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} \right\}^2$$

$$= (x_1 + x_2)^2 + (y_1 + y_2)^2 + (x_1 - x_2)^2 + (y_1 - y_2)^2$$

$$= x_1^2 + 2x_1x_2 + x_2^2 + y_1^2 + 2y_1y_2 + y_2^2 + x_1^2 - 2x_1x_2 + x_2^2 + y_1^2 - 2y_1y_2 + y_2^2$$

$$= 2x_1^2 + 2x_2^2 + 2y_1^2 + 2y_2^2$$

$$= 2x_1^2 + 2y_1^2 + 2x_2^2 + 2y_2^2$$

$$= 2(x_1^2 + y_1^2) + 2(x_2^2 + y_2^2)$$

$$= 2|z_1|^2 + 2|z_2|^2$$

$$= R.H.S$$

$$\therefore L.H.S = R.H.S \text{ (proved)}$$

problem:

Show that the ellipse $|z+3| + |z-3| = 10$ can be expressed in rectangular form as $\frac{x^2}{25} + \frac{y^2}{16} = 1$.

Solution:

Given that

$$|z+3| + |z-3| = 10$$

$$\Rightarrow |x+iy+3| + |x+iy-3| = 10 \quad [\because z = x+iy]$$

$$\Rightarrow |(x+3)+iy| + |(x-3)+iy| = 10$$

$$\Rightarrow \sqrt{(x+3)^2 + y^2} + \sqrt{(x-3)^2 + y^2} = 10$$

$$\Rightarrow \sqrt{(x+3)^2 + y^2} = 10 - \sqrt{(x-3)^2 + y^2}$$

$$\Rightarrow \{\sqrt{(x+3)^2 + y^2}\}^2 = \{10 - \sqrt{(x-3)^2 + y^2}\}^2 \quad \left[\begin{array}{l} \text{Squaring on} \\ \text{both sides} \end{array} \right]$$

$$\Rightarrow (x+3)^2 + y^2 = (10)^2 - 2 \cdot 10 \sqrt{(x-3)^2 + y^2} + \{(x-3)^2 + y^2\}$$

$$\Rightarrow x^2 + 2 \cdot x \cdot 3 + 3^2 + y^2 = 100 - 20 \sqrt{(x-3)^2 + y^2}$$

$$+ (x-3)^2 + y^2$$

$$\Rightarrow x^2 + 6x + 9 + y^2 = 100 - 20 \sqrt{(x-3)^2 + y^2} + x^2 - 6x + 9 + y^2$$

$$\Rightarrow 6x + 6x - 100 = -20 \sqrt{(x-3)^2 + y^2}$$

$$\Rightarrow 12x - 100 = -20 \sqrt{x^2 - 6x + 9 + y^2}$$

$$\Rightarrow 4(3x - 25) = -20 \sqrt{x^2 - 6x + 9 + y^2}$$

$$\Rightarrow 3x - 25 = -5 \sqrt{x^2 - 6x + 9 + y^2} \quad [\text{dividing by 4}]$$

$$\Rightarrow (3x - 25)^2 = \{-5 \sqrt{x^2 - 6x + 9 + y^2}\}^2 \quad [\text{Squaring}]$$

$$\Rightarrow (3x)^2 - 2 \cdot 3x \cdot 25 + (25)^2 = 25 (x^2 - 6x + 9 + y^2)$$

$$\Rightarrow 9x^2 - 150x + 625 = 25x^2 - 150x + 225 + 25y^2$$

$$\Rightarrow 9x^2 - 25x^2 - 25y^2 = 225 - 625$$

$$\Rightarrow -16x^2 - 25y^2 = -400$$

$$\Rightarrow -(16x^2 + 25y^2) = -400$$

$$\Rightarrow 16x^2 + 25y^2 = 400$$

$$\Rightarrow \frac{x^2}{25} + \frac{y^2}{16} = 1 \quad [\text{dividing by } 400]$$

(proved)

problem:

If $z = x + iy$, find the locus represented by $3|z-1| = 2|z-2|$

Solution:

Given that $3|z-1| = 2|z-2|$

$$\Rightarrow 3|x+iy-1| = 2|x+iy-2|$$

$$\Rightarrow 3|(x-1)+iy| = 2|(x-2)+iy|$$

$$\Rightarrow 3\sqrt{(x-1)^2 + y^2} = 2\sqrt{(x-2)^2 + y^2}$$

$$\Rightarrow \{3\sqrt{(x-1)^2 + y^2}\}^2 = \{2\sqrt{(x-2)^2 + y^2}\}^2 \quad [\text{squaring}]$$

$$\Rightarrow 9\{(x-1)^2 + y^2\} = 4\{(x-2)^2 + y^2\}$$

$$\Rightarrow 9(x^2 - 2x + 1 + y^2) = 4(x^2 - 4x + 4 + y^2)$$

$$\Rightarrow 9x^2 - 18x + 9 + 9y^2 = 4x^2 - 16x + 16 + 4y^2$$

$$\Rightarrow 9x^2 - 4x^2 + 9y^2 - 4y^2 = -16x + 18x + 16 - 9$$

$$\Rightarrow 5x^2 + 5y^2 = 2x + 7$$

$$\Rightarrow 5(x^2 + y^2) = 2x + 7 \quad \text{which represent a circle.}$$